Retail Expansion Opportunity Analysis Using Geospatial Analytics

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Software Used: ArcGIS Pro
Data Sources: WorldPop, OpenStreetMap, Delhi Ward Boundaries

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1. Project Overview

Retail chains rely on location intelligence to determine where new stores should be opened. Poor site selection can lead to low customer traffic, competition overlap, and inefficient logistics.

This project applies **geospatial analytics and network analysis** to identify potential **retail expansion locations in Delhi, India**. By combining population data, existing retail store locations, road accessibility, and service area analysis, the study identifies **high-demand areas that are currently underserved by retail stores**.

The final output highlights **optimal candidate locations for retail expansion** based on spatial demand and accessibility.

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2. Business Problem

A retail organization wants to expand its operations in Delhi. However, opening stores without understanding demand distribution and competitor coverage may lead to poor investment decisions.

Key questions addressed in this analysis:

- Which areas in Delhi have **high population demand**?
- Which areas are already **served by existing retail stores**?
- Where are the **gaps in retail accessibility**?
- Which locations represent **optimal expansion opportunities**?

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3. Data Sources

The project integrates multiple geospatial datasets:

Population Data

WorldPop Gridded Population Dataset

Used to estimate population density across Delhi and identify high-demand areas.

Retail Locations

OpenStreetMap Points of Interest

Retail-related locations such as supermarkets, malls, and stores were extracted to represent existing competitors.

Road Network

OpenStreetMap Road Data

Used to build a road network dataset for performing service area and accessibility analysis.

Administrative Boundaries

Delhi Ward Boundaries

Used to aggregate population data and define the study area.

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4. Tools and Technologies

The analysis was performed using:

- **ArcGIS Pro** – spatial analysis and network modeling
- **Network Analyst** – service area analysis
- **OpenStreetMap datasets** – road network and retail points
- **WorldPop** – population density data

Techniques used include:

- Spatial overlay analysis
- Network service area modeling
- Retail market gap analysis
- Location suitability scoring

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5. Methodology

The analysis followed a structured geospatial workflow.

Step 1 – Population Analysis

WorldPop gridded population data was processed and aggregated to ward boundaries to estimate **population density across Delhi**.

High-population wards were identified to represent areas with strong potential demand for retail services.

Step 2 – Retail Competitor Mapping

OpenStreetMap point data was filtered to extract **retail-related locations** such as supermarkets and malls. These locations represent existing retail competitors in the market.

Step 3 – Road Network Preparation

OpenStreetMap road data was cleaned and imported into a **network dataset** to enable routing and accessibility analysis.

Step 4 – Network Service Area Analysis

Service areas were generated around existing retail stores using network analysis.

Three accessibility thresholds were used:

- **2 km** – local accessibility
- **3 km** – neighborhood retail access
- **5 km** – broader urban retail coverage

These service areas represent zones currently served by retail stores.

Step 5 – Identification of Retail Coverage Gaps

High-population areas were compared with existing retail service areas to identify locations that are **outside current retail coverage**.

These areas represent **potential market gaps**.

Step 6 – Suitability Analysis

Candidate locations were evaluated based on several criteria:

• Population density
• Proximity to road infrastructure
• Commercial land use suitability
• Distance from existing retail coverage

Each location was assigned a **suitability score** to prioritize expansion opportunities.

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6. Results

The analysis revealed several **high-population zones in Delhi that fall outside the 5 km retail service coverage area**.

These locations represent strong candidates for retail expansion because they:

• Have **high population demand**
• Are **accessible through road networks**
• Are **not currently served by nearby retail stores**

The final map highlights **recommended locations for potential retail expansion**.

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7. Key Insights

• Retail coverage is concentrated in central commercial districts.
• Peripheral urban areas show **higher population density but lower retail accessibility**.
• Network analysis revealed **several underserved residential zones** within Delhi.

Expanding retail stores in these areas could significantly improve market coverage.

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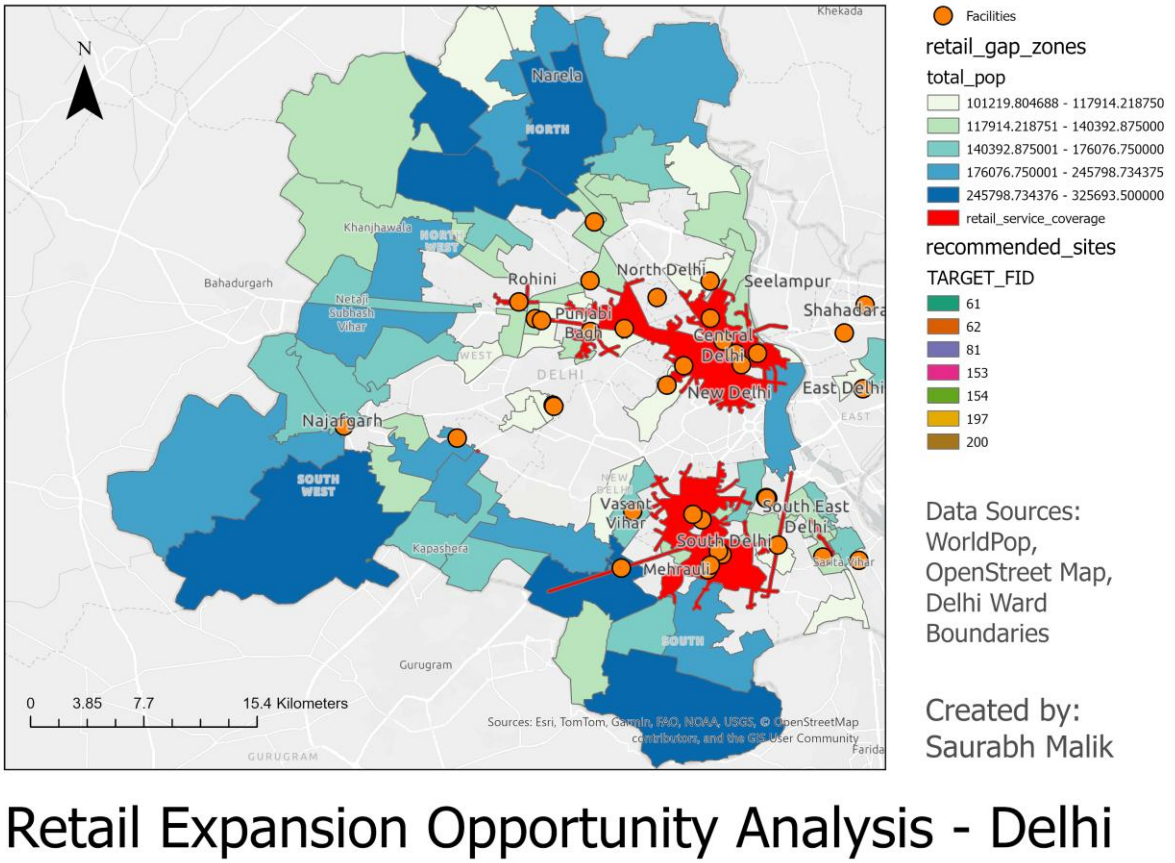
8. Final Map



This map displays population demand, existing retail coverage, and recommended expansion zones.

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9. Conclusion

This project demonstrates how **geospatial analytics can support strategic retail expansion decisions**.

By integrating population demand, transportation networks, and existing competitor coverage, spatial analysis can identify **optimal locations for new stores**.

The methodology applied in this project can be adapted for various industries including retail, logistics, and urban planning.

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10. Future Improvements

Future enhancements could include:

- Demand forecasting using machine learning
- Spatial clustering analysis for market segmentation
- Integration of demographic and income data
- Real-time mobility and traffic analysis

These additions would further refine retail expansion recommendations.

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11. Author

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Geospatial Analytics | Data Science | Location Intelligence